

Short-Term Photovoltaic Forecasting under Data Scarcity using Leave-One-Plant-Out Evaluation

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Abstract—Newly commissioned photovoltaic (PV) plants lack the local generation history that conventional forecasting models require. The *Cold-Start* generation problem translates into operational and financial risk for grid operators and small distributed generators. We investigate whether a single *Cross-Site* deep-learning model, pretrained on many consolidated plants, can forecast a brand-new plant it has never observed, and how it compares against strictly local baselines. We benchmark eight configurations such as global Temporal Fusion Transformer (TFT), Informer, N-HiTS, LSTM and a global XGBoost against strictly local XGBoost, LSTM and N-HiTS. Evaluation is performed under a Leave-One-Plant-Out (LOPO) protocol that hides the target PV plant during training and performs inference only using the training plants. The testing protocol spans 24-hour and autoregressive 7-day horizons over 94 Chilean plants (2014–2024), with point (RMSE, sMAPE, rRMSE) and probabilistic (90 % interval coverage, pinball loss) metrics, stratified by season. Because the comparison is strictly paired on a shared hourly grid, we assess it with a Friedman omnibus test followed by pairwise Wilcoxon signed-rank tests with Holm correction, and we report errors normalised by installed capacity (NRMSE) alongside rRMSE. Local XGBoost is the most accurate configuration (13.2 % median NRMSE, weekly roll-out) and beats the best global model (global XGBoost, 17.1 %) by a significant margin ($p < 10^{-7}$, winning on 77 of 94 plants). No global configuration significantly outperforms its local counterpart at matched architecture: local N-HiTS beats global N-HiTS ($p < 10^{-10}$) and the LSTM pair is statistically indistinguishable ($p = 1.00$, 49/45 plants). The rejection of the global-superiority hypothesis is therefore unambiguous. The value of the Cross-Site approach lies instead in availability: it is the only paradigm that yields coherent, calibrated forecasts from the first connected hour, where no local model can exist.

Index Terms—photovoltaic forecasting, transfer learning, cold-start, deep learning, Temporal Fusion Transformer, probabilistic forecasting, leave-one-plant-out.

I. INTRODUCTION

The accelerated deployment of solar photovoltaic (PV) generation confronts grid operators with a recurrent operational barrier: a plant that has just been connected to the network has no operating history from which to learn its production dynamics. Classical PV forecasting pipelines train a dedicated model per plant on months or years of local data [1]; applied to a freshly commissioned facility they degrade sharply or cannot be trained at all. This is the *Cold-Start* problem, borrowed from recommender systems [2] where reliable inference is required exactly when site-specific evidence is absent. The

resulting uncertainty inflates balancing costs and financial exposure for system operators and, acutely, for the small distributed generators that dominate the recent capacity additions.

A promising solution is to abandon the one-model-per-plant assumption. Instead of learning in isolation, a single *global* model can be trained across many plants simultaneously, absorbing shared atmospheric-to-power patterns and transferring them to unseen sites [3], [4]. Modern attention-based sequence models such as the Temporal Fusion Transformer (TFT) [5], Informer [6] as well as efficient hierarchical alternatives such as N-HiTS [7] make this *Cross-Site* strategy technically feasible on multivariate, exogenous-rich PV data. Despite rapid advances in foundation models for time series [8]–[11], a gap remains in the PV forecasting literature: the performance of global Cross-Site models under a strictly zero-shot protocol has not been contrasted against strong local baselines evaluated under an identical protocol.

Much of the literature reports favourable global-transfer results but rarely combines three demanding conditions at once: (i) a target-plant-blind evaluation with no local fine-tuning, (ii) a direct contrast against strong local baselines under an identical protocol, and (iii) a probabilistic characterization of the forecasts. We close this gap. Our contributions are:

- A reproducible LOPO benchmark of **eight configurations** (five global, three strictly local) over **94 Chilean PV plants**, with formally verified zero-shot integrity (no feature is a function of the target’s own generation).
- A **Synthetic Cold-Start** mechanism that initializes the inference context from a regional efficiency prior computed exclusively on training plants, enabling inference with zero target history without leakage.
- A **point-and-probabilistic** evaluation stratified by season and by evaluation window, reported with capacity-normalised errors and **paired statistical tests** (Friedman + Wilcoxon–Holm) rather than marginal medians alone. It exposes (a) the statistically significant rejection of the global-superiority hypothesis, (b) the absence of any architecture at which global transfer significantly beats its local counterpart, (c) winter as the hardest regime, and (d) a systematic miscalibration of the deep networks’ intervals under transfer.

II. RELATED WORK

PV forecasting. Deep networks consistently surpass classical statistical methods for short-term PV power [12], [13], and on real solar datasets attention-based models improve over classical machine learning at the cost of higher data and compute demand [14]. The TFT has been applied to day-ahead PV forecasting with quantile outputs and interpretable covariate fusion [15]. A recurrent limitation across these works is the assumption of sufficient local history for training.

Global vs. local forecasting. Montero-Manso and Hyndman [3] propose a single global model fitted across related series. The authors show that global model can generalize at least as well as many local models, given enough cross-series signal. For intermittent series, however, the balance between local and global models is not settled [16], and simple baselines often rival transformers on standard benchmarks [17].

Transfer learning for PV. Bak et al. [18] pretrain on large multi-plant corpora and fine-tune on the first records of a new site, attenuating error during the first weeks of operation. Foundation-model approaches push toward pure zero-shot irradiance transfer [19]. Our study differs by enforcing a strict zero-shot regime *and* comparing against strong local baselines.

Zero-shot and few-shot forecasting. A recent family of pretrained time-series foundation models targets forecasting on series unseen during training. Chronos tokenises series and reuses language-model architectures [8]; TimeGPT offers a zero-shot forecasting API [9]; Moirai addresses arbitrary frequencies and covariates with a masked encoder [10]; Lag-Llama builds a decoder-only probabilistic forecaster over lagged features [20]; and TimesFM shows that a decoder-only model pretrained on a large corpus is competitive zero-shot [11]. These models are trained across heterogeneous domains, whereas our global models are trained within a single domain (PV) and transferred across *sites*. Few-shot adaptation, in which a handful of local observations is used to fine-tune, sits between our strict zero-shot setting and the mature local baselines, and is the natural next step rather than a competing condition here.

Uncertainty calibration. Interval calibration is rarely verified under distribution shift. Conformal prediction provides distribution-free coverage guarantees under exchangeability [21]; conformalized quantile regression combines quantile models with conformal correction [22]; EnbPI extends the idea to time series without exchangeability [23]; and adaptive conformal inference retains coverage under distribution shift by updating the miscoverage level online [24]. Recalibration of regression outputs has also been addressed post hoc [25]. We do not apply these methods here; we *measure* whether calibration survives zero-shot transfer, and find that it does not, which motivates them as future work.

Probabilistic forecasting. Quantile and distributional outputs are standard for decision-relevant energy forecasting [26], [27]. We adopt 90 % prediction intervals and evaluate coverage and pinball loss, but crucially examine whether calibration itself survives the zero-shot transfer.

III. METHODOLOGY

A. Dataset and Medallion ETL

The corpus comprises hourly generation from 94 PV plants of the Chilean National Electric System (2014–2024), dynamic exogenous meteorology (global horizontal irradiance, temperature, humidity) from the CR2 network, and static geographic attributes (macro-zone, installed capacity). The plants span the country’s steep solar gradient and are grouped into four macro-zones from the Atacama desert in the north to the south; this grouping is the static key over which the regional prior is aggregated. A Medallion pipeline (landing → bronze → silver → gold, Fig. 1) parses the raw meteorological and generation CSVs, standardizes them to a canonical (*unique_id*, *ds*, *y*) schema, joins them into a single `silver_unified` table ($\approx 4.37\text{M}$ hourly rows), engineers cyclic calendar features, and imputes *only* exogenous variables (each flagged with an `is_imputed` dummy). Generation *y* is never imputed: missing hours remain missing, and metrics score real observations only. A numeric, label-encoded `silver_dl` view feeds the deep networks, while the gold layer materializes one held-out test partition per plant for the LOPO evaluation. A single orchestrator runs the pipeline behind a pre-run test gate with idempotent completion markers.

B. LOPO Protocol and Zero-Shot Integrity

Evaluation follows a Leave-One-Plant-Out design. For each target plant, the global models are trained on the full 2014–2024 history of the remaining $N-1$ plants; the target is *completely invisible* during training. Zero-shot integrity is a hard constraint: no predictor may be a function of the target’s own *y*. This is enforced by automated tests and was decisive—an early efficiency prior computed per-row from the target ($\text{corr} = 1.0$ with *y*) produced a fake $\sim 0.8\%$ rRMSE that collapsed to the honest $\sim 56\%$ once removed.

C. Synthetic Cold-Start Context

Deep models require a context window to initialize inference, which a new plant cannot provide. We seed it synthetically:

$$\hat{y}_{\text{seed}}(t) = \text{capacity} \cdot PR_{\text{regional}} \cdot s(t), \quad (1)$$

where $s(t)$ is a deterministic solar profile and PR_{regional} is a performance ratio aggregated by macro-zone and season *over the training plants only*. Only known or forecastable exogenous weather and static geography use the target’s real values; the generation context is entirely synthetic (Fig. 2).

D. Models and Probabilistic Output

We evaluate five global configurations—TFT [5], Informer [6], N-HITS [7], LSTM [28] and a global gradient-boosted XGBoost [29]—and three strictly local baselines (XGBoost, LSTM, N-HITS) trained only on the target’s own history with all evaluation windows excluded. The four deep architectures span complementary inductive biases: the TFT

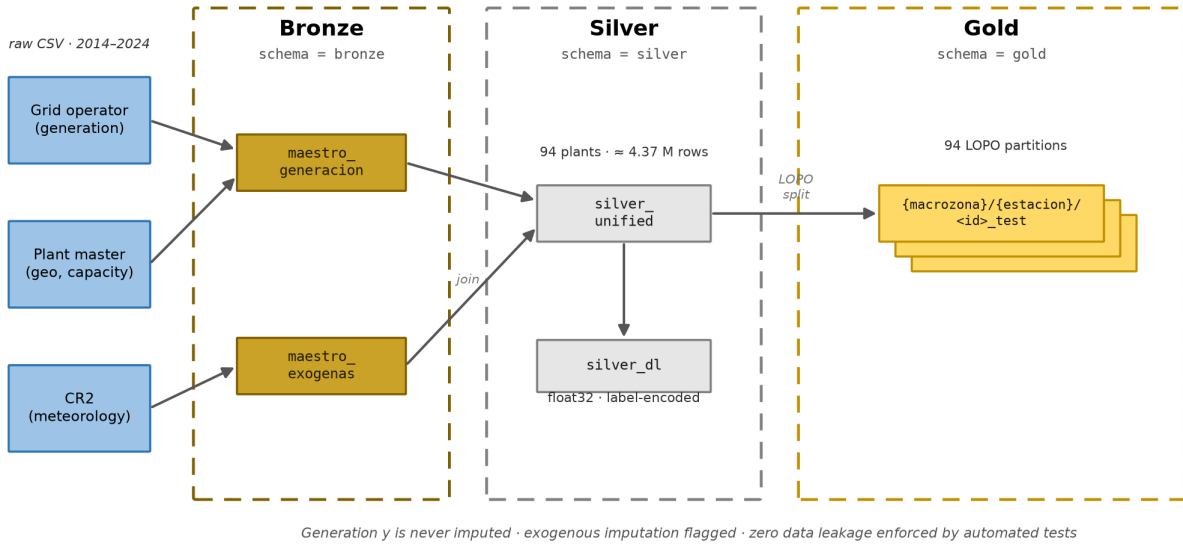


Fig. 1. Medallion ETL data-lineage. Source systems feed two Bronze master tables, joined into `silver_unified` and its numeric `silver_dl` view; the gold layer holds one LOPO test partition per plant. Generation is never imputed, and zero leakage is enforced by automated tests.

combines variable-selection networks with interpretable multi-head attention and gating; the Informer replaces full attention with a ProbSparse mechanism for long sequences; N-HiTS performs hierarchical multi-rate interpolation; and the LSTM is a strong recurrent baseline. Deep models are trained with NeuralForecast in `fp32` on a 6GB GPU under an epoch-based step budget capped per strategy. Every model emits P5/P50/P95 quantiles (90% intervals), and predictions are physically constrained ($\hat{y} \geq 0$; forced to zero at night).

Data separation and hyperparameter selection. The hold-out in LOPO is *across plants*, not across time: for a given target, the training set is the complete 2014–2024 history of the other $N-1$ plants, and the test set is the target’s evaluation window. Early stopping uses the last 168 hours of each training series as a validation split, so validation is temporally posterior to the data used for the gradient updates of that series. Hyperparameters are searched once per architecture (Optuna, on a fixed development subset of training plants) and reused across all folds. This reuse is a deliberate compute trade-off and constitutes *parameter-level* information sharing across folds. We bound its effect rather than assert it away: the search space is small (learning rate, hidden size, number of layers), the same configuration is applied to every fold so no fold is individually tuned on its own target, and the ranking we report is driven by differences far larger than the spread observed across the tuning trials. It remains a limitation, and a per-fold search would be the strictly correct protocol.

Temporal causality of the local baselines. The local baselines are trained on the target plant’s full history *with all evaluation windows removed*—Raw, Operational and the

four Seasonal windows—rather than on data strictly preceding each window. Lags that would reach into a removed window are masked, so no evaluation target leaks in as a feature; but the training set does include hours chronologically *after* the window being predicted. This is a deliberate choice: the local baseline is meant to represent the mature model an operator would eventually hold once the plant has accumulated history, which is the strongest possible opponent for the Cross-Site approach. It is nonetheless an optimistic setting for the local side. The global side carries a mirror-image advantage: LOPO hides the target plant but not the calendar, so a global model does observe the same hours at neighbouring plants of the same macro-zone. Both biases favour the conclusion being tested against—the local side is flattered and it wins, while the global side is flattered and it still loses—so they reinforce rather than undermine the rejection reported in Section IV-E. A strictly causal rolling-origin protocol is left as future work.

E. Evaluation Windows and Metrics

Two horizons are scored: *day1* (24h direct) and *rollout7d* (7 autoregressive days feeding back the predicted median). Because early series often contain a *commissioning ramp* of null generation, we report a window sensitivity analysis (Fig. 2): *Raw* (from the first recorded hour, ramp included), *Operational* (from first sustained production) and four *Seasonal* windows starting in each season, drawn from the plant’s own later history. Point metrics are RMSE, sMAPE and two normalisations of the RMSE. The first, rRMSE, divides by the window’s mean hourly generation; because that mean includes null night hours it inflates the index, and values above 100%

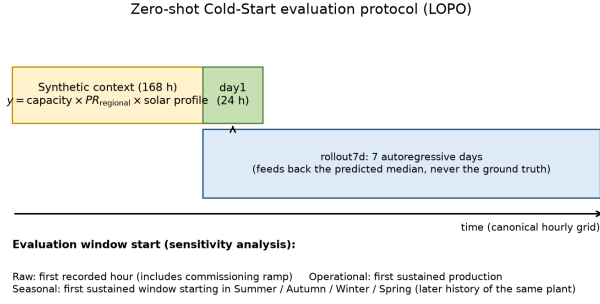


Fig. 2. Zero-shot Cold-Start evaluation protocol. A 168-hour synthetic context seeds the inference; the model then predicts the *day1* (24 h) and *rollout7d* horizons. The window start defines the Raw/Operational/Seasonal sensitivity analysis.

are expected rather than anomalous. We retain rRMSE for continuity with the thesis this work condenses, but we add

$$\text{NRMSE} = \frac{\text{RMSE}}{P_{\text{inst}}} \times 100, \quad (2)$$

normalised by the plant’s installed capacity P_{inst} , which is bounded, directly interpretable as a percentage of nameplate power, and comparable across plants of very different size (0.3–200 MW here). NRMSE is also the metric on which the statistical tests are run, because it is defined for every plant, whereas rRMSE is undefined when a window’s mean generation is zero. Probabilistic quality uses 90 % coverage and pinball loss; we report *diurnal* coverage (hours with $y > 0$) as the honest measure, since the physical night-zero forcing yields trivial nocturnal hits. Reported cells are medians over plants, with interquartile ranges given for the headline metric.

What the global–local comparison is for. The two families are not evaluated under equivalent information: a global model sees no data at all from the target plant, whereas a local baseline sees that plant’s own history. The comparison is therefore not a like-for-like contest but a deliberate test of a practical question—*does knowledge transferred from other plants substitute for the plant’s own history?* We report it at two levels: against the strongest available local baseline, which bounds what the operator gives up by using transfer, and at *matched architecture* (global vs. local LSTM, global vs. local N-HITS), which isolates the effect of the training regime from that of the model family.

IV. RESULTS

Results correspond to the *total-scale* run: LOPO over the 94 target plants, global models trained on the complete $N-1$ history. Table I reports the cold-start window for all 94 plants; Fig. 3 visualizes the weekly roll-out ranking.

A. Point Accuracy

Three readings emerge from the weekly roll-out. First, **local XGBoost** is the most accurate configuration (13.2 % median NRMSE, i.e. an RMSE of 13 % of nameplate power), followed

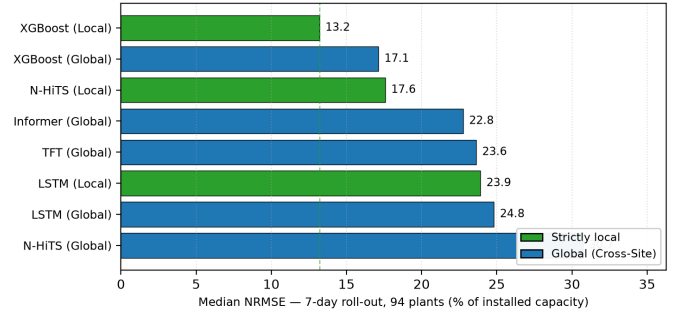


Fig. 3. Median NRMSE (7-day roll-out, cold-start window, 94 plants) per configuration, as a percentage of installed capacity. Local models in green, global Cross-Site models in blue; the dashed line marks the best local baseline.

by **global XGBoost** (17.1 %), the best Cross-Site model. Second, the deep networks form a clearly worse group (22.8–30.7 % NRMSE) in which **N-HITS global** is the weakest; the ordering is consistent with the caution that attention can be over-provisioned relative to simpler baselines [17]. Third, the degradation from *day1* to the weekly roll-out is moderate across all families, indicating that autoregressive median feedback does not destabilize the forecasts.

The two normalisations also illustrate why the choice matters. Ranked by rRMSE, global LSTM (108.6 %) appears clearly ahead of local LSTM (128.8 %); ranked by NRMSE the order reverses (24.8 % vs. 23.9 %). Marginal medians of a ratio whose denominator varies by plant are not a safe basis for ranking, which is precisely why we turn to paired tests.

B. Commissioning-Ramp Window Sensitivity

Only 33 of the 94 plants exhibit a commissioning ramp longer than 24 h; for the rest the Raw and Operational windows coincide. On the aggregate median the two definitions differ only moderately, but on plants with a ramp the Raw window is distorted: starting the evaluation over weeks of null generation drives the window mean toward zero and makes rRMSE undefined. Reporting both definitions is therefore deliberate—Raw is faithful to the historical record, whereas Operational yields interpretable metrics for a plant already in commercial operation and is the window used for the headline comparison.

C. Seasonal Sensitivity

For every plant we build Cold-Start windows starting in each of the four seasons, taken from its own post-commissioning history (independent of the actual connection season). Winter is **systematically the hardest season for every architecture** (Table II): in the weekly roll-out, global XGBoost rises from 51.8 % (summer) to 133.5 % (winter) and local XGBoost from 40.5 % to 114.8 %. Greater cloud variability and lower mean winter irradiance weaken the deterministic solar signal and amplify the relative weight of errors, with a direct operational implication: the forecast uncertainty of a newly connected plant depends strongly on the season of its commissioning.

TABLE I

MEDIAN PERFORMANCE BY CONFIGURATION OVER ALL **94 PLANTS**, COLD-START WINDOW, SORTED BY NRMSE. FOR THE 33 PLANTS WITH A COMMISSIONING RAMP LONGER THAN 24 H THE OPERATIONAL WINDOW IS USED; FOR THE REMAINING 61, WHOSE SUSTAINED PRODUCTION STARTS WITHIN 24 H OF THE FIRST RECORD, THE RAW WINDOW ALREADY BEGINS IN SUSTAINED PRODUCTION AND IS USED INSTEAD. rRMSE NORMALISES BY THE WINDOW'S MEAN GENERATION, NRMSE BY INSTALLED CAPACITY; IQR IS THE INTERQUARTILE RANGE OF THE ROLL-OUT NRMSE ACROSS PLANTS. COV₉₀ IS DIURNAL COVERAGE OF THE 90 % INTERVAL AND PINBALL P95 IS IN MWh. BEST VALUE PER COLUMN IN **BOLD**.

Model	Scope	rRMSE (%)		NRMSE (% cap.)		IQR _{NRMSE}	Cov ₉₀	Pinball
		day1	roll7d	day1	roll7d			
XGBoost	Local	45.6	59.9	9.5	13.2	[8.7, 17.9]	0.86	0.035
XGBoost	Global	64.0	75.1	15.6	17.1	[13.6, 21.6]	0.97	0.092
N-HiTS	Local	62.3	85.2	14.2	17.6	[13.4, 22.7]	0.72	0.096
Informer	Global	96.6	106.5	21.0	22.8	[19.3, 31.6]	0.64	0.115
TFT	Global	105.1	112.7	22.0	23.6	[18.9, 29.0]	0.38	0.158
LSTM	Local	102.6	128.8	16.3	23.9	[16.1, 33.8]	0.72	0.235
LSTM	Global	99.7	108.6	22.6	24.8	[20.9, 31.8]	0.53	0.159
N-HiTS	Global	129.8	148.2	28.6	30.7	[21.6, 42.6]	0.37	0.372

Observed vs. predicted generation with 90% prediction intervals (plant: canesa solar i)

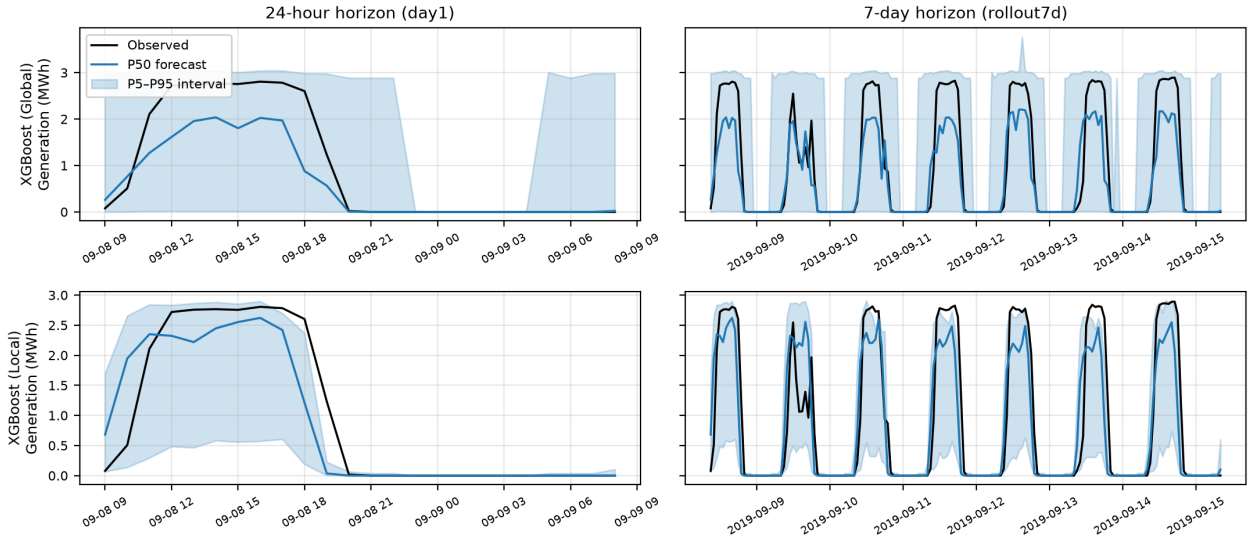


Fig. 4. Observed vs. predicted generation with 90 % prediction intervals for the same plant and window, at the 24-hour (left) and 7-day (right) horizons. Top: global XGBoost, whose median under-forecasts the peaks and whose bands are far too wide (over-coverage). Bottom: local XGBoost, which tracks the daily cycle closely with near-nominal bands.

TABLE II

MEDIAN rRMSE (%) OF THE 7-DAY ROLL-OUT BY SEASON IN WHICH THE EVALUATION WINDOW STARTS, FOR ALL EIGHT CONFIGURATIONS.

Model	Scope	Summer	Autumn	Winter	Spring
XGBoost	Local	40.5	35.9	114.8	101.5
XGBoost	Global	51.8	53.7	133.5	106.1
N-HiTS	Local	66.0	74.0	143.5	118.1
N-HiTS	Global	114.0	130.3	188.7	148.9
LSTM	Local	94.1	139.6	166.1	155.1
LSTM	Global	93.9	97.0	186.3	136.2
TFT	Global	87.7	102.0	168.8	130.9
Informer	Global	87.9	92.9	172.0	137.0
plants		94	92	93	94

D. Probabilistic Calibration

No paradigm calibrates its intervals ideally under the Cold-Start protocol, with deviations in opposite directions from the 90 % nominal (Table I, Cov₉₀). Evaluated on productive hours, the global deep networks are clearly **under-calibrated** (global LSTM 0.53, Informer 0.64, TFT 0.38, N-HiTS 0.37): their bands are too narrow. Global XGBoost, at the opposite extreme, **over-covers** (0.97)—bands so wide they rarely fail but lose operational value, a defect symmetric to the networks', not a virtue. The closest to nominal is **local XGBoost** (0.86), whose superiority is confirmed by a much smaller pinball P95 (0.035 vs. 0.092 MWh for global). The networks' behaviour follows from the design: each series' local scaler is fit on the synthetic context, whose variance (an idealized solar profile) is below the real plant variability, producing

TABLE III
KEY CONTRASTS ON ROLL-OUT NRMSE OVER 94 PLANTS. PAIRED WILCOXON SIGNED-RANK WITH HOLM CORRECTION; “WINS” COUNTS PLANTS ON WHICH THE FIRST CONFIGURATION IS BETTER. A DASH MEANS THE DIFFERENCE IS NOT SIGNIFICANT AT $\alpha = 0.05$.

Contrast	Better	Δ	Wins	p
XGB: local vs. global	local	3.7	77/17	$< 10^{-4}$
LSTM: global vs. local	—	0.3	49/45	1.000
N-HiTS: global vs. local	local	13.8	6/88	$< 10^{-4}$
Global: XGB vs. LSTM	XGB global	9.0	75/19	$< 10^{-4}$

narrow bands; the local deep baselines, which see real context, improve (LSTM local and N-HiTS local both 0.72). Post-hoc recalibration—conformalized quantile regression [22], time-series conformal intervals [23] or adaptive conformal inference under shift [24]—is a clear line of future work.

E. Statistical Significance

Because every configuration is evaluated on the same plants over the same canonical hourly grid, the design is fully paired and balanced (94 plants $\times$ 8 configurations), which is the setting in which rank-based comparisons over multiple datasets apply [30]. We run a Friedman omnibus test on the roll-out NRMSE followed by all 28 pairwise Wilcoxon signed-rank tests with Holm correction.

The omnibus test rejects the null of equal performance decisively ($\chi^2 = 260.7$, $p < 10^{-50}$). Mean ranks (1 = best) order the configurations as local XGBoost 1.80, local N-HiTS 3.32, global XGBoost 3.34, TFT 5.10, Informer 5.15, local LSTM 5.30, global LSTM 5.41, global N-HiTS 6.59. Counting per-plant winners gives the same picture: local XGBoost is best on 60 of 94 plants, global XGBoost on 15, local N-HiTS on 10, and the remaining four configurations on 9 plants combined.

Table III reports the contrasts that bear on the hypothesis. Local XGBoost beats global XGBoost on 77 of 94 plants ($p < 10^{-7}$), so the headline rejection is significant and not an artifact of aggregation. At matched architecture the picture is now unambiguous: local N-HiTS beats global N-HiTS on 88 of 94 plants ($p < 10^{-10}$), and the LSTM pair is **statistically indistinguishable** (median paired difference 0.3 NRMSE points, 49/45 plants, $p = 1.00$ after correction). *There is no architecture at which global transfer significantly outperforms its local counterpart.*

This result corrects a reading that marginal medians invite. On the earlier operational subset, global LSTM showed a much better median rRMSE than local LSTM (118.2 vs. 146.9%), which suggested a genuine advantage for transfer at matched architecture. The paired difference tells a different story: the median of the per-plant differences is essentially zero and slightly favours the local model. The gap between the two medians was driven by a minority of plants on which the local LSTM fails badly, not by a systematic advantage of the global one. The hypothesis is therefore rejected without the nuance we previously claimed.

The operational value of the global model is consequently not accuracy but *availability*: the local paradigm needs months of accumulated data, whereas the global model forecasts useably from the first connected hour, a regime in which no local competitor exists.

Fig. 4 makes the same contrast visible at the series level for a representative plant: the global model reproduces the diurnal shape from weather and static geography alone, but systematically under-shoots the peaks and hedges with very wide intervals, whereas the local model tracks both the level and the shape with much tighter bands.

V. DISCUSSION

The results show that Cross-Site transfer retains value in a pure cold start scenario. Several factors may affect the global models’ performance and define the roadmap. The global model ran strictly zero-shot while the local baselines that beat it observe real site data; a two-stage pretrain-then-finetune scheme should close much of the gap as history accrues. Also, the models treat PV as a continuous signal, whereas it is intrinsically intermittent (deterministic night zeros, stochastic cloud gaps); explicit intermittency masks in the attention mechanism could redirect representational capacity to the productive signal. *Corpus scale and compute*: 94 plants and a 6 GB training budget may under-serve the higher-capacity TFT/Informer, offering an alternative explanation for their being outperformed by simpler models. Finally, *Interval recalibration* under transfer (conformal methods) is needed before operational deployment.

Beyond the benchmark, four takeaways can be generalized to any cold-start forecasting study. (1) Commissioning ramps of null generation distort relative metrics unless the window is defined from sustained production, which motivated our Raw/Operational split. (2) Winter is systematically the hardest regime, doubling or tripling summer errors across all architectures. (3) Probabilistic calibration does not transfer to the deep networks under a synthetic context, whereas quantile gradient boosting calibrates well—a critical distinction for electricity-market use. (4) Target integrity requires never imputing missing generation and is essential to metric honesty, in contrast to the common practice of zero-filling.

Limitations. The study is zero-shot by design, with no fine-tuning arm, so it bounds what transfer achieves without any local adaptation rather than what a few-shot scheme could reach. The local baselines are trained non-causally with respect to their evaluation window (Section III-E), which favours them; a rolling-origin protocol would settle the margin exactly. Hyperparameters are shared across folds. The rRMSE normalisation inflates absolute magnitudes and is meaningful only in relative comparison, which is why NRMSE and the paired tests carry the conclusions. Operational MLOps deployment is out of scope.

VI. CONCLUSION

We benchmarked global Cross-Site transfer against strictly local baselines for cold-start PV forecasting under a target-

blind LOPO protocol over 94 Chilean plants, and assessed the comparison with paired statistical tests rather than marginal medians alone. The global-superiority hypothesis is rejected, and the rejection is statistically significant: a local XGBoost beats the best global configuration on 77 of 94 plants ($p < 10^{-7}$), and no architecture shows a significant advantage for global transfer over its local counterpart. The value of the Cross-Site paradigm is therefore not accuracy but availability: in the pure cold-start regime it is the only approach that produces coherent, immediately usable forecasts, because no local model can be trained at all. Winter is the hardest season, and the deep networks' probabilistic intervals require recalibration under transfer. The contribution is not to crown an absolute winner but to quantify rigorously the trade-off between *immediate availability* and *asymptotic accuracy* that governs paradigm choice during the commissioning of a new solar plant.

REPRODUCIBILITY

The medallion ETL, the LOPO split and all eight configurations are driven by a single orchestrator behind an automated test gate that halts the run on any failure; every metric is computed only over real observations on a shared canonical hourly grid, so the eight models are compared in a strictly paired fashion. Idempotent completion markers make long runs resumable, and the regional prior is recomputed inside each fold to preserve zero-shot integrity. The source code, the per-plant metric tables underlying every figure and table in this paper, and the scripts that compute the statistical tests are available from the authors upon request.

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REFERENCES

- [1] J. Antonanzas, N. Osorio, R. Escobar, R. Urraca, F. J. Martinez-de Pison, and F. Antonanzas-Torres, "Review of photovoltaic power forecasting," *Solar Energy*, vol. 136, pp. 78–111, 2016.
- [2] A. I. Schein, A. Popescul, L. H. Ungar, and D. M. Pennock, "Methods and metrics for cold-start recommendations," in *Proc. 25th Annual Int. ACM SIGIR Conf. on Research and Development in Information Retrieval*, 2002, pp. 253–260.
- [3] P. Montero-Manso and R. J. Hyndman, "Principles and algorithms for forecasting groups of time series: Locality and globality," *Int. J. Forecasting*, vol. 37, no. 4, pp. 1632–1653, 2021.
- [4] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowledge and Data Engineering*, vol. 22, no. 10, pp. 1345–1359, 2010.
- [5] B. Lim, S. Ö. Ank, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *Int. J. Forecasting*, vol. 37, no. 4, pp. 1748–1764, 2021.
- [6] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *Proc. AAAI Conf. on Artificial Intelligence*, vol. 35, 2021, pp. 11 106–11 115.
- [7] C. Challu, K. G. Olivares, B. N. Oreshkin, F. Garza, M. Mergenthaler-Canseco, and A. Dubrawski, "N-HITS: Neural hierarchical interpolation for time series forecasting," in *Proc. AAAI Conf. on Artificial Intelligence*, vol. 37, 2023, pp. 6989–6997.
- [8] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. Pineda Arango, S. Kapoor, J. Zschiesner, D. C. Maddix, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang, "Chronos: Learning the language of time series," *arXiv preprint arXiv:2403.07815*, 2024.
- [9] A. Garza and M. Mergenthaler-Canseco, "TimeGPT-1," *arXiv preprint arXiv:2310.03589*, 2023.
- [10] G. Woo, C. Liu, A. Kumar, C. Xiong, S. Savarese, and D. Sahoo, "Unified training of universal time series forecasting transformers," in *Proc. 41st Int. Conf. on Machine Learning (ICML)*, 2024.
- [11] A. Das, W. Kong, R. Sen, and Y. Zhou, "A decoder-only foundation model for time-series forecasting," in *Proc. 41st Int. Conf. on Machine Learning (ICML)*, 2024.
- [12] A. Mellit, A. M. Pavan, and V. Lughi, "Deep learning neural networks for short-term photovoltaic power forecasting," *Renewable Energy*, vol. 172, pp. 276–288, 2021.
- [13] P. Kumari and D. Toshniwal, "Deep learning models for solar irradiance forecasting: A comprehensive review," *J. Cleaner Production*, vol. 318, p. 128566, 2021.
- [14] Z. Arslan, M. E. Sahin, and H. Ulutas, "Comparative evaluation of machine learning and transformer-based deep learning for solar energy forecasting on a real-world dataset," *Arabian J. for Science and Engineering*, vol. 51, pp. 6833–6863, 2025.
- [15] M. López Santos, X. García-Santiago, F. Echevarría Camarero, G. Blázquez Gil, and P. Carrasco Ortega, "Application of temporal fusion transformer for day-ahead pv power forecasting," *Energies*, vol. 15, no. 14, p. 5232, 2022.
- [16] S. Damato, N. Rubattu, D. Azzimonti, and G. Corani, "Intermittent time series forecasting: local vs global models," *arXiv preprint arXiv:2601.14031*, 2026.
- [17] A. Zeng, M. Chen, L. Zhang, and Q. Xu, "Are transformers effective for time series forecasting?" in *Proc. AAAI Conf. on Artificial Intelligence*, vol. 37, 2023, pp. 11 121–11 128.
- [18] S. Bak, S. Choi, D. Yang, D. Kim, H. Rho, and K. Lee, "Transfer learning for photovoltaic power forecasting across regions using large-scale datasets," *IEEE Access*, vol. 13, pp. 136 175–136 190, 2025.
- [19] A. Mishra, T. Ravindra, S. Iyengar, S. Kalyanaraman, and P. Kumaraguru, "SPIRIT: Short-term prediction of solar irradiance for zero-shot transfer learning using foundation models," *arXiv preprint arXiv:2502.10307*, 2025.
- [20] K. Rasul, A. Ashok, A. R. Williams, A. Khorasani, G. Adamopoulos, R. Bhagwatkar, M. Biloš, H. Ghonia, N. Hassen, A. Schneider, S. Garg, A. Drouin, N. Chapados, Y. Nevmyvaka, and I. Rish, "Lag-Llama: Towards foundation models for probabilistic time series forecasting," *arXiv preprint arXiv:2310.08278*, 2023.
- [21] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York: Springer, 2005.
- [22] Y. Romano, E. Patterson, and E. J. Candès, "Conformalized quantile regression," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.
- [23] C. Xu and Y. Xie, "Conformal prediction interval for dynamic time-series," in *Proc. 38th Int. Conf. on Machine Learning (ICML)*, 2021.
- [24] I. Gibbs and E. J. Candès, "Adaptive conformal inference under distribution shift," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021.
- [25] V. Kuleshov, N. Fenner, and S. Ermon, "Accurate uncertainties for deep learning using calibrated regression," in *Proc. 35th Int. Conf. on Machine Learning (ICML)*, 2018.
- [26] D. Salinas, V. Flunkert, J. Gasthaus, and L. Stella, "DeepAR: Probabilistic forecasting with autoregressive recurrent networks," *Int. J. Forecasting*, vol. 36, no. 3, pp. 1181–1191, 2020.
- [27] K. Benidis, S. S. Rangapuram, V. Flunkert *et al.*, "Deep learning for time series forecasting: Tutorial and literature survey," *ACM Computing Surveys*, vol. 55, no. 6, pp. 1–36, 2022.
- [28] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [29] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [30] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.